MANISH PAL

EMBEDDED SYSTEMS | FIRMWARE DEVELOPMENT | HARDWARE DEBUGGING

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PROFESSIONAL SUMMARY

Electronics and Communication Engineering undergraduate focused on embedded systems, firmware development, microcontroller integration, and hardware debugging. Brings approximately 3 years of hands-on electronics troubleshooting experience and a two-month Embedded Systems Development internship. Has independently built ESP32-based IoT systems and microcontroller prototypes, including a state-level winning industrial monitoring project. Seeking internship and opportunities in embedded firmware, embedded hardware, and hardware validation.

TECHNICAL SKILLS

Microcontrollers: ESP32, STM32, Arduino, Raspberry Pi Pico ,Respberry pi 3 Model A+

Programming: Embedded C (working), C++ (basic), C

Embedded development: PlatformIO, STM32CubeMX, GPIO, sensor/actuator integration, I^2C, SPI, UART

Hardware and testing: PCB fault diagnosis, circuit tracing, soldering, component replacement, power-supply troubleshooting, multimeter and oscilloscope

Design and tools: KiCad schematic/PCB design, Git, GitHub, Linux

Additional: FreeRTOS/RTOS (working knowledge), Embedded Linux (working knowledge), Wi-Fi/IoT communication, web dashboards

SELECTED PROJECTS

NEXUS-6 — Industrial Monitoring Network

- I developed a six-room monitoring system using ESP32 client nodes and a host dashboard.
- Integrated sensor monitoring, Wi-Fi communication, and a local web dashboard; the system is working and has been demonstrated.
- State-level 1st-prize project at Sewa Sankalp Innovation Challenge 2026; the related IoT Server work was also recognized at district level.

Smart Energy Monitor — Commercial/Industrial Energy Monitoring

- Built an individual energy-monitoring system covering voltage, current, power, frequency, and energy-unit measurements.
- Designed an integrated experience involving a display, phone dashboard, website, alerts, remote controls, usage summaries, and next-month bill estimation.

GateX — Proximity-Based Automated Gate System

- Built and demonstrated an individual gate-automation prototype with proximity-based open/close behavior and OLED status feedback.
- Implemented buzzer notification, mute-button control, and red/green status indicators; developed with productization in mind.

STM32 Oscilloscope

- Developed an STM32-based oscilloscope prototype; final measurement specifications and public repository are not yet documented.

AI Smart Attendance System

- Developed a working prototype for an attendance system; describe as a prototype rather than a completed production deployment.

STM32 TinyML Classifier

- Currently in the initial development stage; included as an in-progress project, not as a completed implementation.

EXPERIENCE

Electronics Technician | Practical Electronics Repair and Troubleshooting • *Approximately 3 years*

- Diagnosed PCB faults and performed circuit tracing, component replacement, and soldering.
- Troubleshoting , systems , and power supplies using practical fault-isolation techniques.

Embedded Systems Development Intern | Dewan VS Group of Engineering & Technology • *Summer internship, 2 months*

- Built two embedded-system projects and gained hands-on development experience across more than five microcontroller platforms.
- Received an internship certificate; internship year to be confirmed.

EDUCATION

B.Tech, Electronics and Communication Engineering (ECE) — Dewan VS Group of Engineering & Technology, Meerut
Expected graduation: 2028
Class XII, UP Board — SD Inter College, Sadar, Meerut | 65.6% | 2025
Class X, UP Board — SVN Inter College | 77.16% | 2023

RESEARCH, AWARDS & LEADERSHIP

- IEEE conference research paper published; title, conference name, year, and DOI/link to be added.
- State-level 1st prize — Sewa Sankalp Innovation Challenge 2026 (NEXUS-6).
- District-level 1st prize winning recognition for the related IoT Server project; official award title to be confirmed.
- College-level winning project — Smart Energy Monitor at Technophilia; smart energy monitoring.
- Workshop organizer, mentor, and trainer for an embedded systems workshop supported by an IEEE Student Chapter and a college club.
- Additional certificates in volunteering/coordination, quizzes and competitions, basic computing, and hands-on PCB and embedded-systems workshops.

LANGUAGES

Hindi (fluent) • English (working proficiency)